Econ 306 Summer Session I

You may not use any class materials to answer the question or consult with any classmates. If AI is used to answer any of the questions you will recieve a 0 and be reported to the Honor Council. Write you answer on a separate piece of paper and then take a photo/scan your paper and upload to ELMS at the end of the quiz. You will have 20 minutes to complete the quiz and then 10 minutes to upload your answers. The quiz is out of 13 points. Please show your work as partial credit may be given for correct work.

Quiz 3

Two firms produce a homogeneous product. Let p denote the product's price. The output level of firm 1 is denoted by q_1, and the output level of firm 2 by q_2. The aggregate industry output is denoted by $Q = q_1 + q_2$. The industry inverse demand curve for this product is given by

$$p = 200 - 4Q.$$

Each firm has a constant average and marginal cost $AC = MC = 20$.

1. What is the best response function for firm 1? (2 points)
2. What is the Cournot-equilibrium output of firm 1? (2 points)
3. What is the Cournot-equilibrium market output ($q_1 + q_2$) (2 points)?
4. What is the Cournot-equilibrium market price? (2 points)
5. How much profit does firm 1 make in the Cournot equilibrium? (2 points)
6. Suppose the two firms decide to collude by forming a cartel that chooses output cooperatively. They agree to each produce half of the monopoly output in the market. What output does each firm produce? (3 points)